Functional Specifications of Converter

1. Introduction

Converter is a program for converting files from an Indian language to its WX format in English and vice versa. The input files can be in SSF (Shakti Standard Format) or simple UTF/text files. This makes use of Conversion modules of the corresponding Indian language in which mappings are written for their UTF characters. Thus, this program provides a common format output for all the Indian languages. In SSF file; TKN, LEX, VIB, NAME and HEAD fields will be converted. In case of text files, the whole words will be converted.

2. Input-Output Specifications

2.1 Input file is SSF

Input:

```
((        CAT_      <fs af=TKN_,,,,,,V_,'name='N_'>
TKN_1     CAT_1     <fs af=TKN_1,,,,,,V_1,'head='H_1'>
))
```

Output:

```
((        CAT_      <fs af=TKN_2,,,,,,V_2,'name='N_2'>
TKN_3     CAT_1     <fs af=TKN_3,,,,,,V_3,'head='H_3'>
))
```

2.2 Input file is Text Format

Input:

```
W_1 W_2 W_3
```

Output:

```
W_4 W_5 W_6
```

Examples:

1 UTF – WX Conversion

1.1 Input file is SSF

INPUT:

ADDR_ TKN_ CAT_ OTHR

```
<Sentence id="1">
1     ((     NP    <fs af='लडका,n,m,pl,,d,,' head="लड़के">
1.1   उस     DEM   <fs af='उस,pn,any,sg,3,o,,'>
1.2   लड़के  NN    <fs af='लडका,n,m,pl,,d,ने,' name="लड़के">
1.3   ने     PSP   <fs af='ने,psp,,,,,,'>
      ))
2     ((     NP    <fs af='रादा,n,f,sg,3,,,' head="रादा">
2.1   रादा   NN    <fs af='रादा,n,f,sg,3,,,' name="रादा">
2.2   का     PSP   <fs af='का,n,m,sg,,d,,'>
      ))
3     ((     NP    <fs af='केला,n,m,sg,,d,,' head="केला">
3.1   केला   NN    <fs af='केला,n,m,sg,,d,,' name="केला">
      ))
4     ((     VGF   <fs af='काया,n,f,sg,,d,,' head="काया">
4.1   काया   VM    <fs af='काया,n,f,sg,,d,,' name="काया">
      ))
5     ((     NP    <fs af='ता,n,m,sg,,d,,' head="ता">
5.1   ता     NN    <fs af='ता,n,m,sg,,d,,' name="ता">
      ))
</Sentence>
```

OUTPUT:

```
<Sentence id="1">
1     ((     NP    <fs af='ladakA,n,m,pl,,d,,' head="ladZake">
1.1   usa    DEM   <fs af='usa,pn,any,sg,3,o,,'>
1.2   ladZake      NN    <fs af='ladakA,n,m,pl,,d,ne,'
name="ladZake">
1.3   ne     PSP   <fs af='ne,psp,,,,,,'>
```

```
        ))
2       ((      NP    <fs af='rAxA,n,f,sg,3,,,' head="rAxA">
2.1     rAxA    NN    <fs af='rAxA,n,f,sg,3,,,' name="rAxA">
2.2     kA      PSP   <fs af='kA,n,m,sg,,d,,'>
        ))
3       ((      NP    <fs af='kelA,n,m,sg,,d,,' head="kelA">
3.1     kelA    NN    <fs af='kelA,n,m,sg,,d,,' name="kelA">
        ))
4       ((      VGF   <fs af='kAyA,n,f,sg,,d,,' head="kAyA">
4.1     kAyA    VM    <fs af='kAyA,n,f,sg,,d,,' name="kAyA">
        ))
5       ((      NP    <fs af='wA,n,m,sg,,d,,' head="wA">
5.1     wA      NN    <fs af='wA,n,m,sg,,d,,' name="wA">
        ))
</Sentence>
```

1.2 Input file is Text format

INPUT:

राणा सांगा को हराने के लिए मुगलों ने यहाँ कई बार आक्रमण किया जिनमें कई बार राणा सांगा घायल हुए |
यहाँ का सबसे बड़ा आर्कषा समुद्र का संगम है |
यहाँ पहुंचना भी मुश्किल नहीं है |
पांडिचेरी में स्मारकों की कमी नहीं है |

OUTPUT:

rANA sAMgA ko harAne ke lie mugaloM ne yahAz kaI bAra AkramaNa kiyA jinameM kaI bAra rANA sAMgA GAyala hue .
yahAz kA sabase badZA AkarRaNa samuxra kA saMgama hE .
yahAz pahuMcanA BI muSkila nahIM hE .
pAMdicerI meM smArakoM kI kamI nahIM hE .

2 WX – UTF Conversion

2.1 Input file is SSF

INPUT:

```
ADDR_ TKN_ CAT_      OTHR

<Sentence id="1">
1       ((      NP    <fs af='ladakA,n,m,pl,,d,ne,' head="ladZake">
1.1     kAyA QF     <fs af='kAyA,n,f,sg,,d,,' name="kAyA">
1.2     bacceNN     <fs af='baccA,n,m,p,,0,0,'>
        ))
2       ((      NP    <fs af='kiwAba,n,f,s,,0,0,'>
2.1     kiwAba      NN    <fs af='kiwAba,n,f,s,,0,0,'>
        ))
3       ((      VGF   <fs af='pada,v,m,p,any,,wA_hE,'>
3.1     padawe      VM    <fs af='pada,v,m,p,any,,wA,'>
3.2     .       SYM
        ))
</Sentence>
```

OUTPUT:

```
<Sentence id="1">
1       ((      NP    <fs af='लडका,n,m,pl,,d,ने,' head="लड़के">
1.1     काया    QF    <fs af='काया,n,f,sg,,d,,' name="काया">
1.2     बच्चे   NN    <fs af='बच्चा,n,m,p,,0,0,'>
        ))
2       ((      NP    <fs af='किताब,n,f,s,,0,0,'>
2.1     किताब NN    <fs af='किताब,n,f,s,,0,0,'>
        ))
3       ((      VGF   <fs af='पड,v,m,p,any,,ता_है,'>
3.1     पडते    VM    <fs af='पड,v,m,p,any,,ता,'>
3.2     ।       SYM
        ))
</Sentence>
```

2.2 Input file is Text format

INPUT:

panI jIvatawA ke kAraNa ve hara wUPAna ke bAxa Sahara kI baxaSakla ho cukI Sakla ko pahale se jyAxA suMxara banA xewe hEM .
jahAz ke maMxiroM Ora mahaloM kI camaka ko samaya kI XuMXa BI XuzXalA na sakI hE .
KajUra ke vqkRoM ke AXAra para hI isa jagaha kA nAma padZA KajurAho .
lekina skOYtalEMda kI pahacAna sirPa yahIM waka sImiwa nahIM hE .

OUTPUT:

पनी जीवटता के कारण वे हर तूफान के बाद शहर की बदशक्ल हो चुकी शक्ल को पहले से ज्यादा सुंदर बना देते हैं |
जहाँ के मंदिरों और महलों की चमक को समय की धुंध भी धुँधला न सकी है |
खजूर के वृक्षों के आधार पर ही इस जगह का नाम पड़ा खजुराहो |
लेकिन स्कॉटलैंड की पहचान सिर्फ यहीं तक सीमित नहीं है |

3. Flow Chart of Converter

Flow chart of the Converter will show the over all view of the program control flow from one module to other.

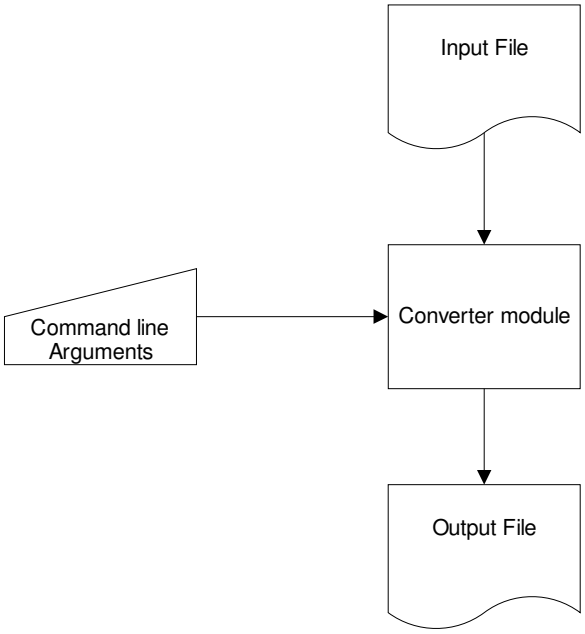


Fig.1- Basic Flow

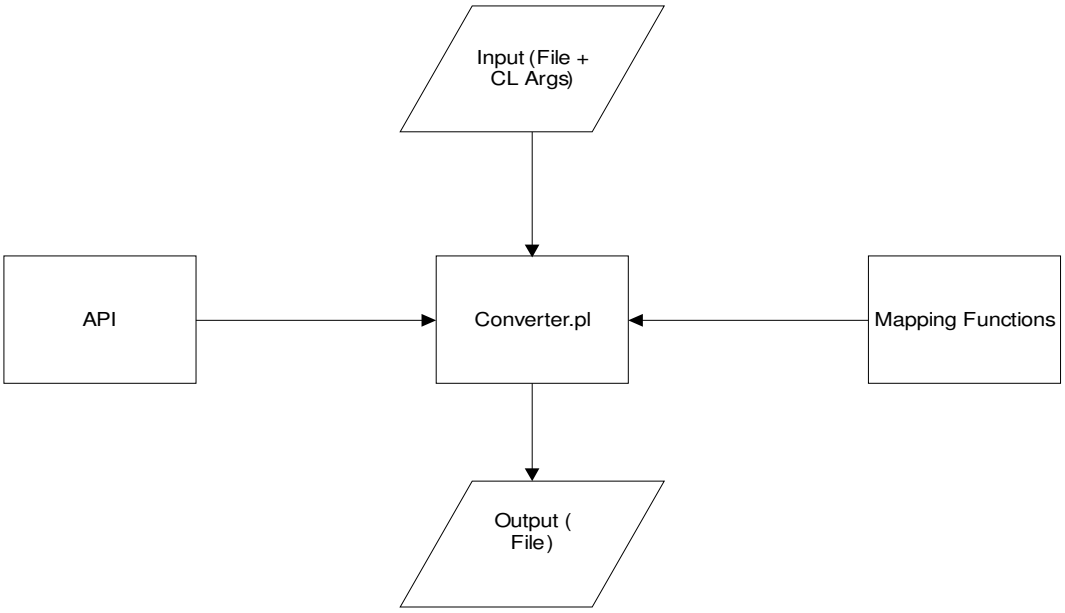


Fig.2-Converter

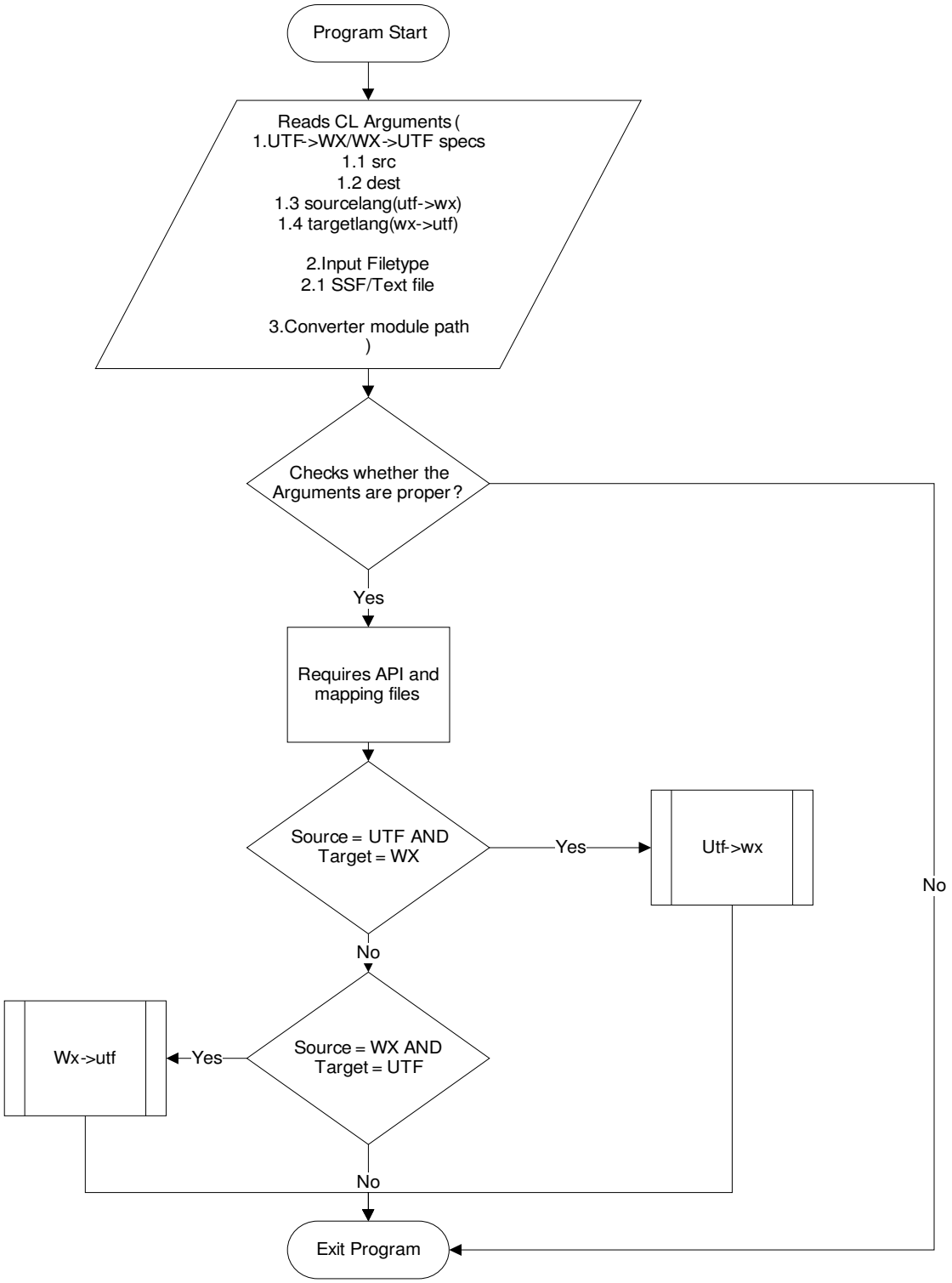


Fig.3-Converter.pl (Main program execution)

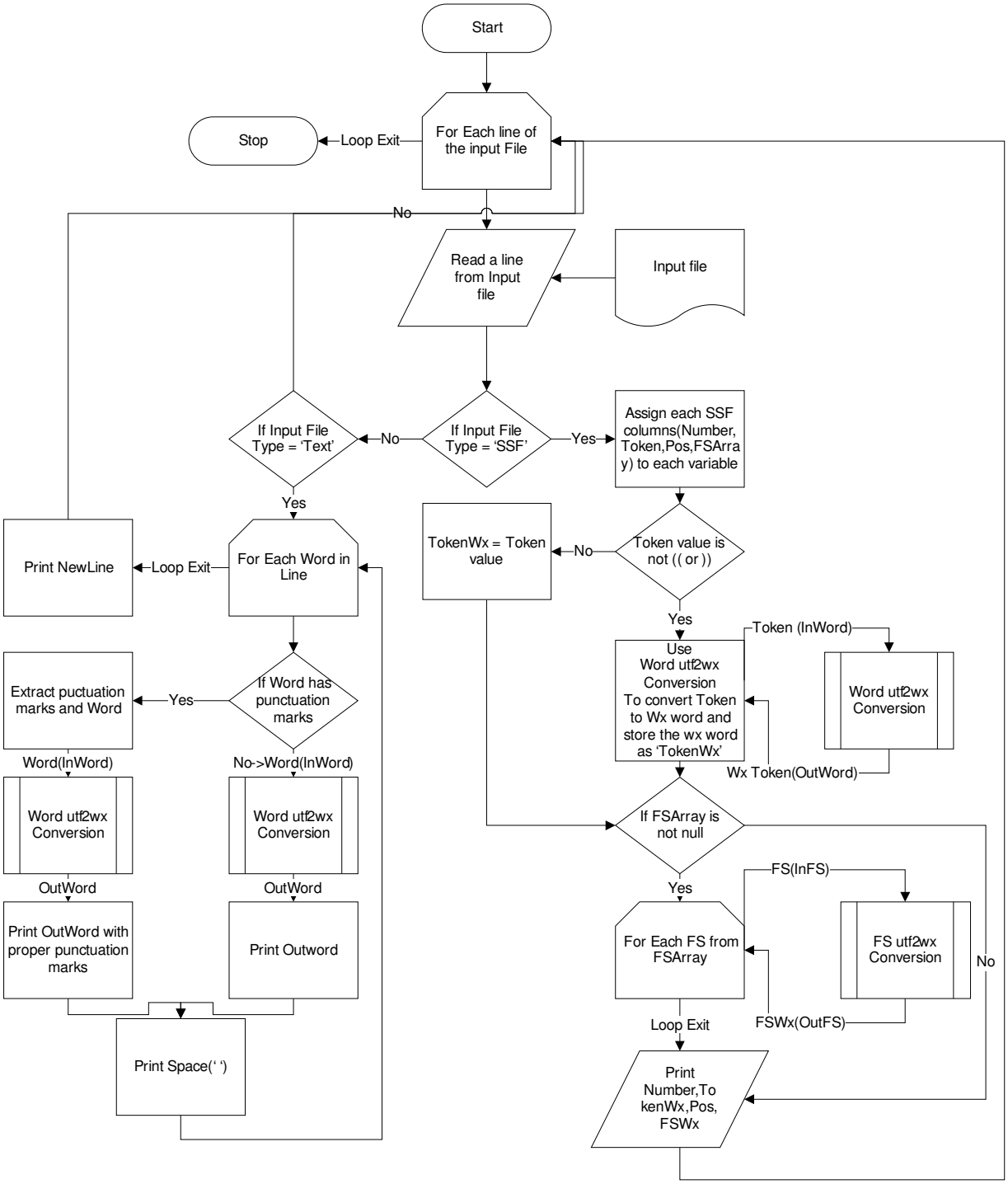


Fig.4-UTF to WX Conversion

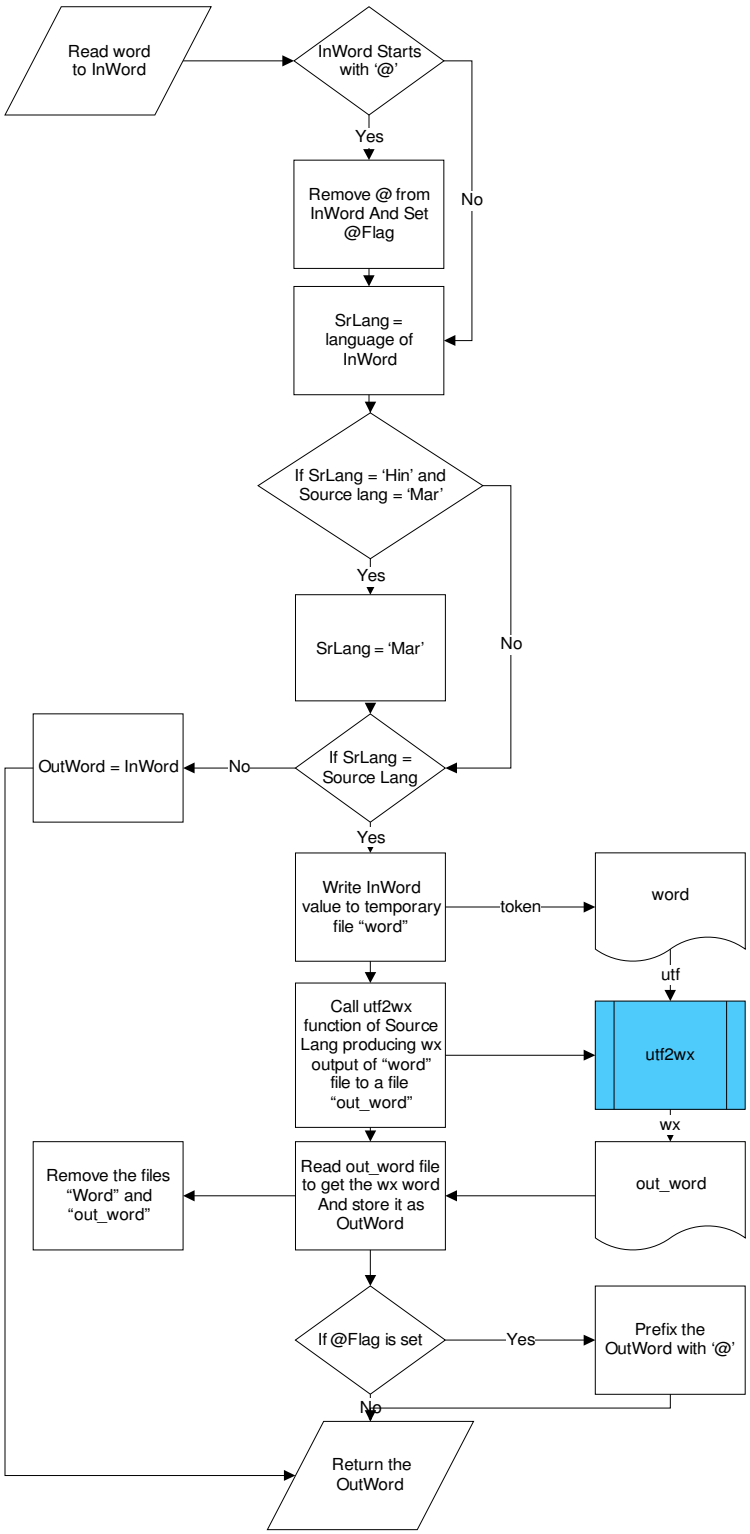


Fig.5-UTF to WX Conversion of a Word

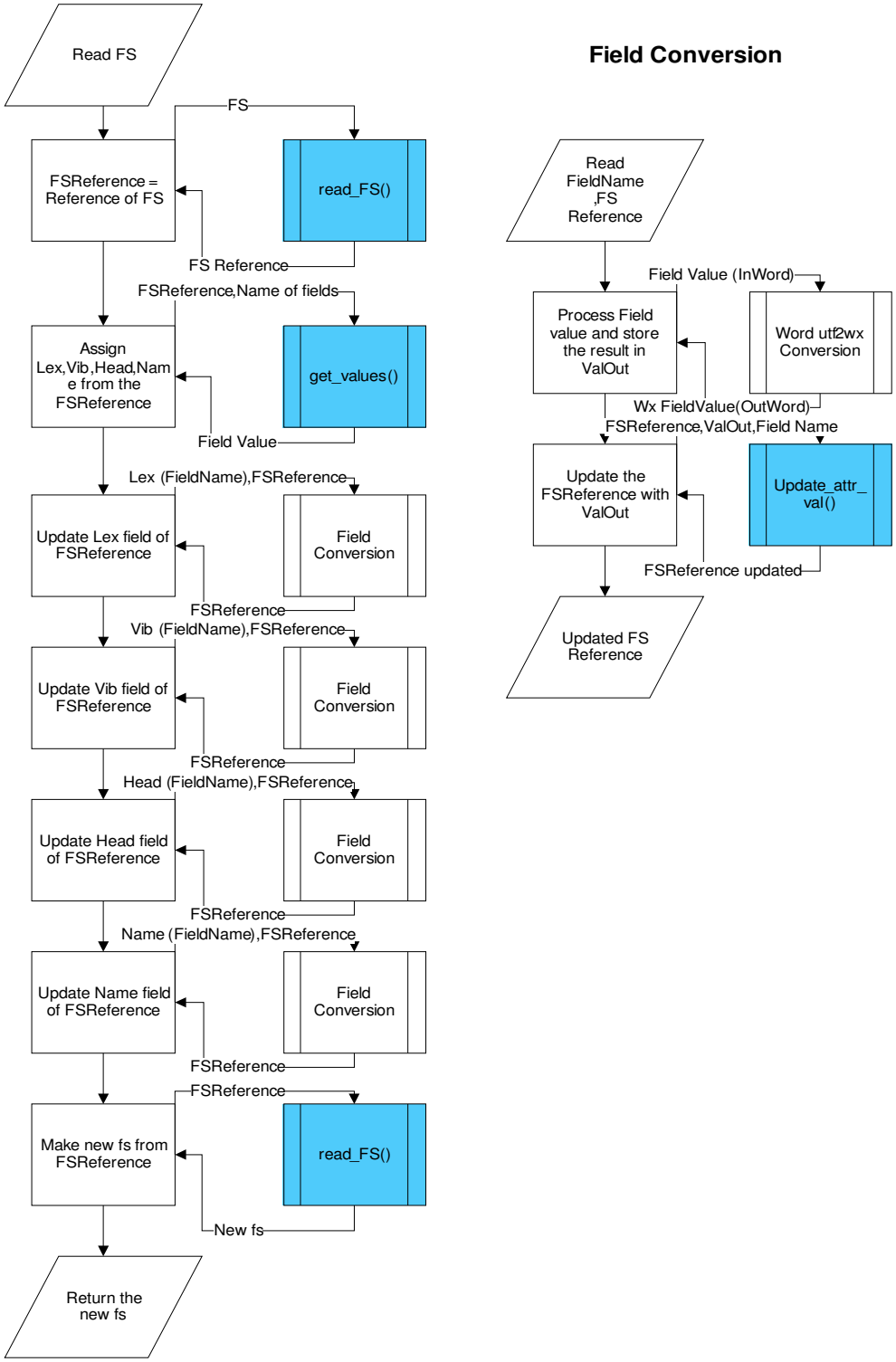


Fig.6-UTF to WX Conversion of Feature Structure

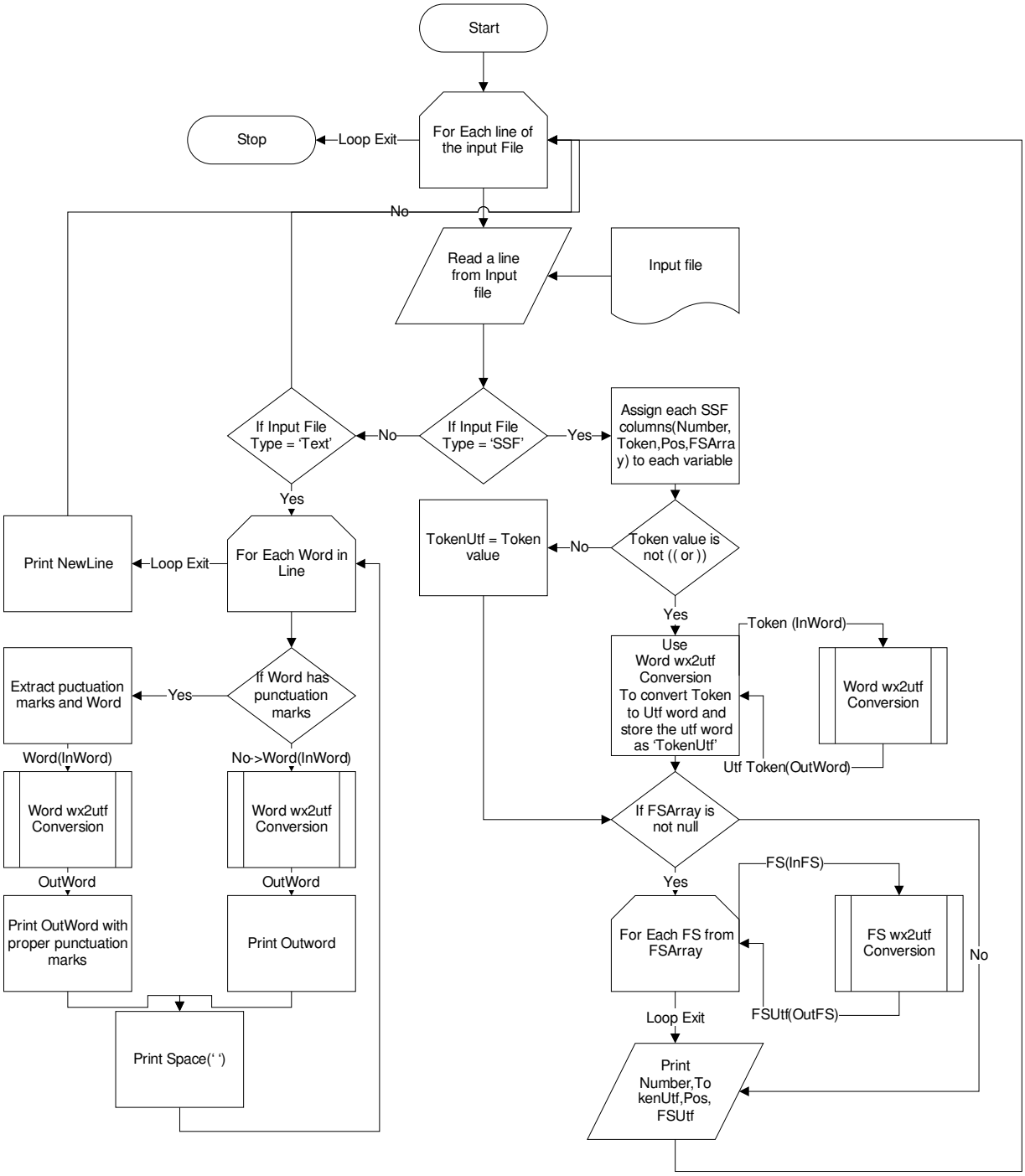


Fig.7-WX to UTF Conversion

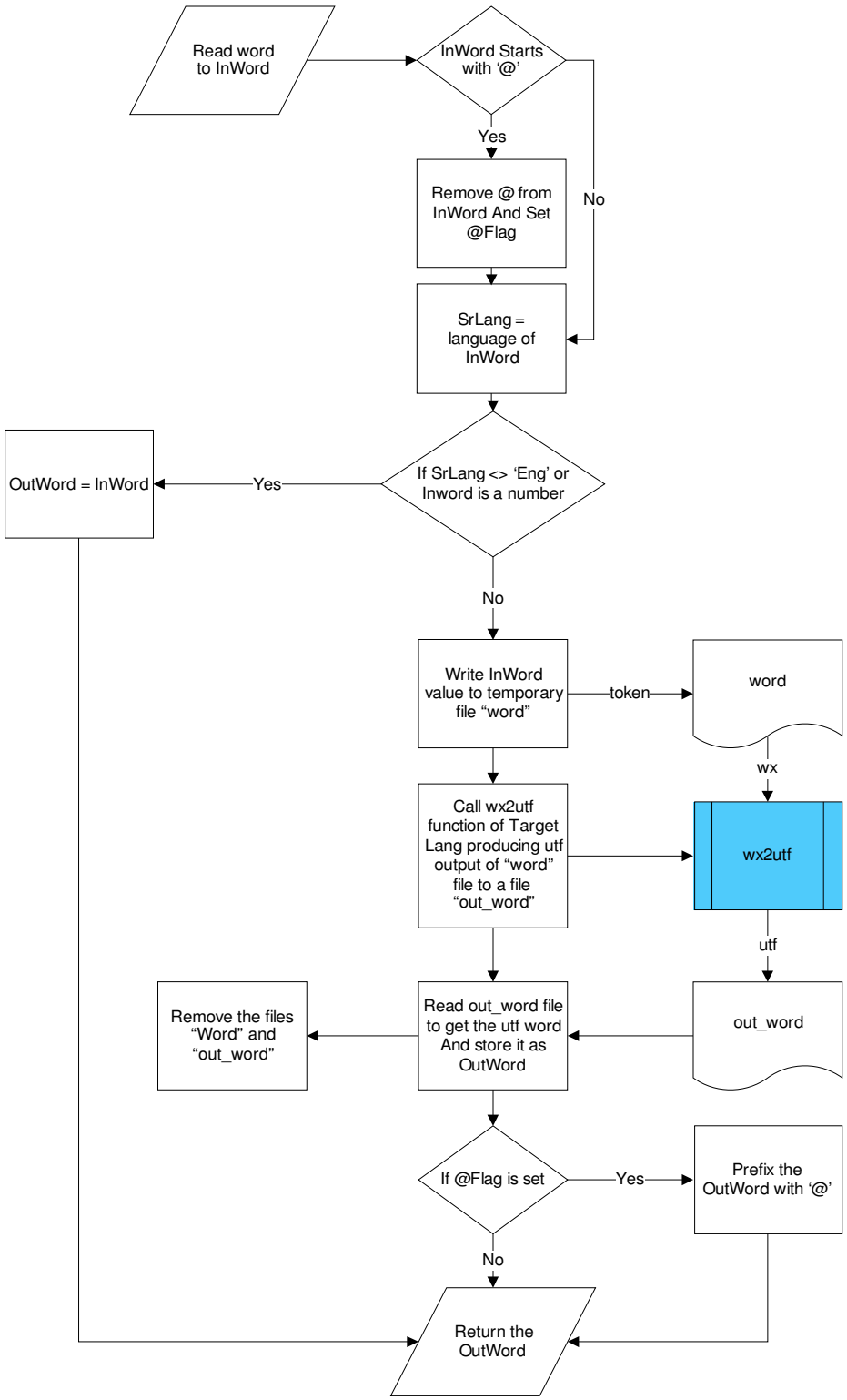


Fig.8-WX to UTF Conversion of a Word

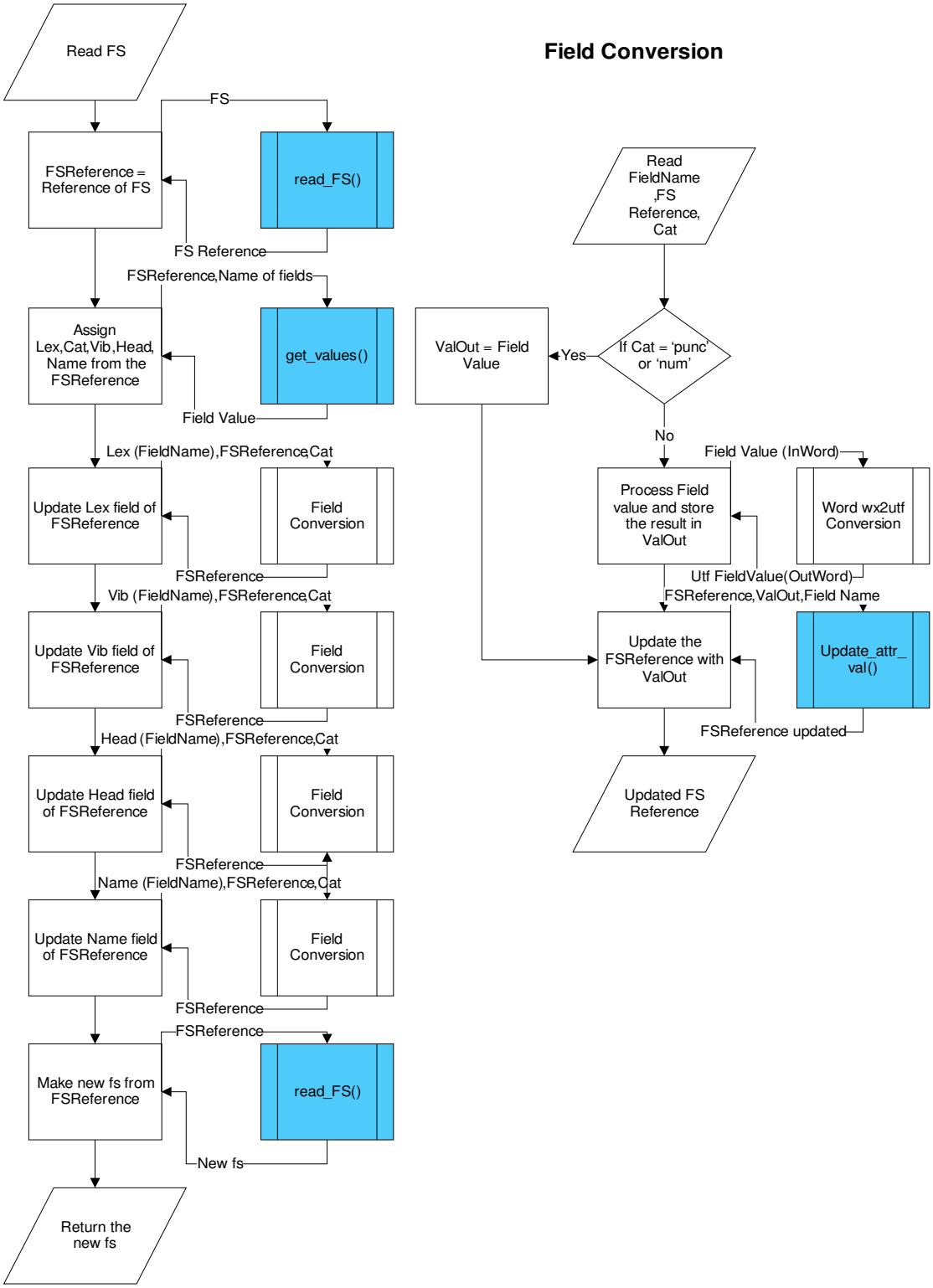


Fig.9-WX to UTF Conversion of Feature Structure

4. Process Description

4.1. Decide the type of Conversion and include the necessary resources

Based on the command line arguments three parameters are identified

1. Type of conversion – UTF to WX or WX to UTF
2. Language of Conversion – Language of UTF file
3. Type of Input File – SSF or UTF/Text file

Include the API and conversion module (based on the language selection)

4.2. Read the Input File

Read the input file line by line. Repeat the next four steps till the EOF.

4.3. Extract the Words to be converted

In case of SSF format extract the TKN and Feature structure using the API functions.

In case of text file read all words in the line separated by space.

4.4. Analyze and Convert the word if required

In case of Feature Structure, it is further divided into LEX, VIB, CAT, NAME and HEAD fields. Except the CAT field all the other fields are converted.

Remove and Store the extra characters (punctuation marks) in the word if it has any. Store the resulting word to a temporary file. Convert that file using the conversion module's mapping function and store the result in another temporary file. Read that file to get the converted word and remove the temporary files. Re-attach the extra characters in the converted word.

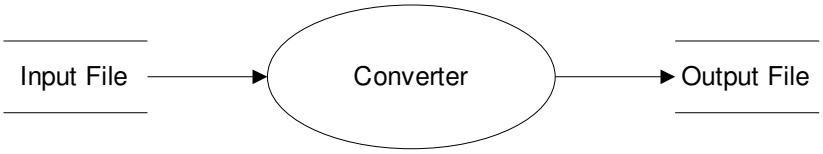
4.5. Reform the line.

Reform the SSF line or simple line based on the file type.

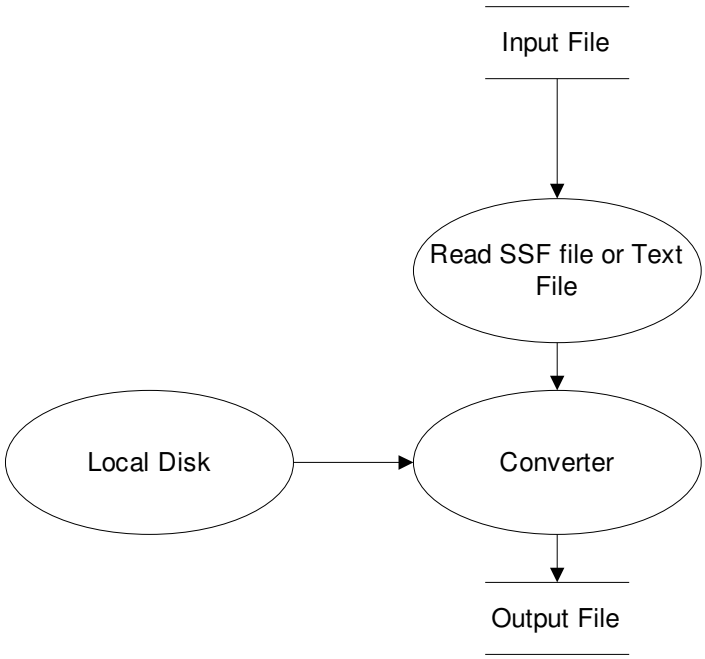
4.6. Print the resulting line to Output File

Print the converted line to the Standard Output.

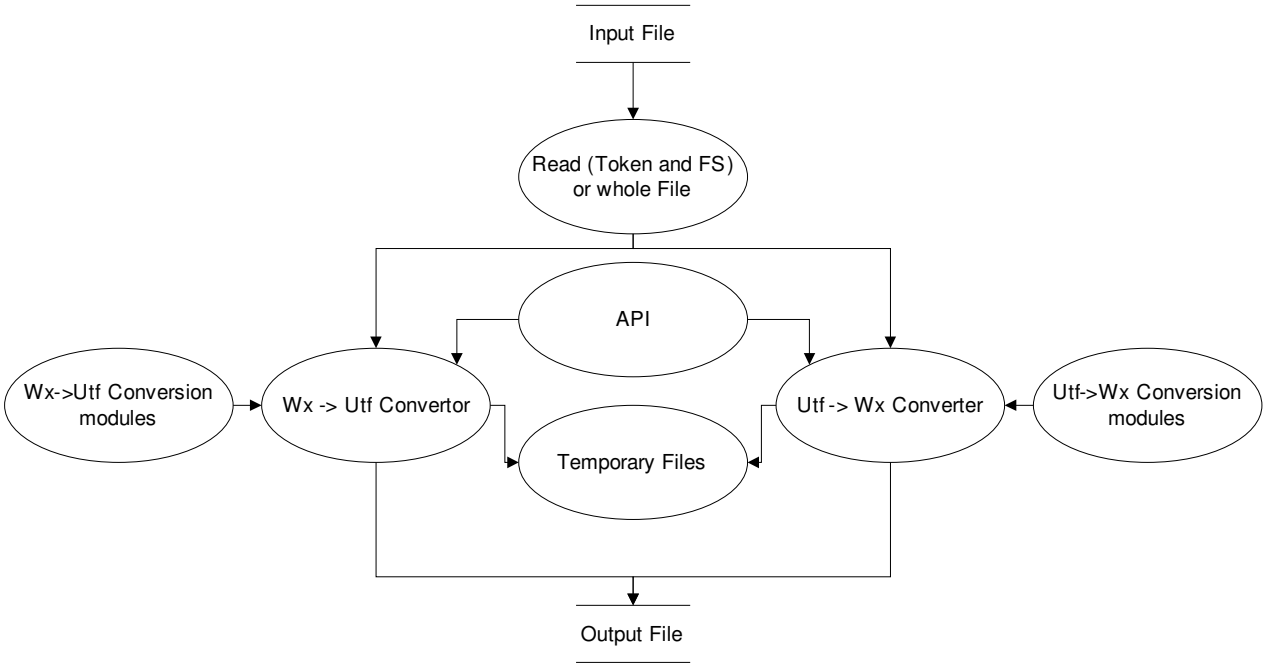
5. Data Flow Diagram



Level 0 DFD



Level 1 DFD


Input File
Read (Token and FS) or whole File
API
Wx->Utf Conversion modules
Wx -> Utf Convertor
Utf -> Wx Converter
Utf->Wx Conversion modules
Temporary Files
Output File

Level 2 DFD